

MODULE-5

Uncertain Knowledge and Reasoning

BCSE306

Artificial Intelligence

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Uncertainty

- When an agent knows enough facts about its environment, the logical plans and actions produces a guaranteed work.
- Unfortunately, *agents never have access to the whole truth about their environment.* Agents act under uncertainty.

Nature of Uncertain Knowledge

- **The Diagnosis** : medicine, automobile repair, or whatever-is a task that almost always involves uncertainty
- Let us try to write rules for **dental diagnosis** using first-order logic, so that we can see how the logical approach breaks down. Consider the following rule:
 - $\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity})$.
- The problem is that this rule is wrong.
- Not all patients with toothaches have cavities; some of them have gum disease, swelling, or one of several other problems
 - $\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity}) \vee \text{Disease}(p, \text{GumDisease}) \vee \text{Disease}(p, \text{Swelling}) \dots$
- to make the rule true, we have to add almost unlimited list of possible causes.
- We could try a causal rule:
 - $\forall p \text{ Disease}(p, \text{Cavity}) \Rightarrow \text{Symptom}(p, \text{Toothache})$.
- But this rule is **also** not right either; not all cavities cause pain
- Toothache and a Cavity are unconnected, so the judgement may go wrong.

Nature of Uncertain Knowledge...

- This is a type of the medical domain, as well as most other judgmental domains: law, business, design, automobile repair, gardening, dating, and so on.
- The agent take action, only a **degree of belief** in the relevant sentences.
- Our main tool for dealing with degrees of belief will be **probability theory**
- **The Probability** assigns to each sentence a numerical degree of belief between 0 and 1.

Probability

- Probabilities are used to compute the truth of given statement, written as numbers between 0 and 1, that describes how likely an event is to occur.
- 0 indicates impossibility and 1 indicates certainly.
 - 1. Tossing a coin 2. Tolling a dice
- Probability based reasoning
 - understanding from knowledge
 - how much of uncertainty present in that event.

Probability

- *Probability provides a way of **summarizing** the uncertainty, that comes from our **laziness** and **ignorance**.*
- Toothache problem - an 80% chance , a probability of 0.8 that the patient has a cavity if he or she has a toothache.
- The 80% summarizes those cases, but both toothache and cavity are unconnected.
- The missing 20% summarizes, all other possible causes of toothache, that we are **too lazy** or **ignorant to confirm** or **deny**.

- Probabilities between 0 and 1 correspond to intermediate degrees of belief in the **truth of the sentence**.
- The sentence itself is in *fact* either **true or false**.
- It is important to note that a **degree of belief** is different from a degree of truth.
- A probability of 0.8 does not mean "80% true" but rather an 80% degree of belief-that is, a fairly strong expectation.
- Thus, probability theory makes the same **ontological commitment** as logic-namely, that facts either do or do not hold in the world.
- Degree of truth, as opposed to degree of belief, is the subject of **fuzzy logic**

- In probability theory, a sentence such as
- "The probability that the patient has a cavity is 0.8",
- is about the agent's beliefs, not directly about the world.
- These percepts create the **evidence**, which are based on probability statements.
- All probability statements must indicate the evidence with respect to that probability is being assessed.
- If an agent receives new percepts, its probability assessments are updated to reflect the new evidence.

Random Variable

- Referring to a "part" of the world, whose "status" is initially unknown
- We will use lowercase for the names of values
 - $P(a) = 1 - P(\neg a)$
- Tossing coin : $P(h) = 1 - P(\neg h) : (0.5 = 1 - 0.5)$
- Rolling dice : $P(n) = 1 - P(\neg n) : (0.16 = 1 - 0.84)$

Types of random variables

- Boolean random variables
 - Cavity domain (true,false), if Cavity = true then cavity, or
 - if Cavity = false then $\neg$ cavity
- Discrete random variables – countable domain
 - *Weather* might be (sunny, rainy, cloudy, snow)
- Continuous random variables – finite set real numbers with equal intervals e.g. interval(0.1)

Atomic events

- The concept of an **atomic event** is useful in understanding the foundations of probability theory.
- It is a **complete specification** of the state of the world about which the agent is uncertain.
- It can be an **assignment of particular values**, to all the variables of which the world is composed

Atomic events...

- Atomic events have some important properties
- They are **mutually exclusive** -at most one can actually be the case.
- The set of all possible atomic events is **exhaustive** – at least one must be the case.
- Any particular atomic event entails the truth or falsehood of every proposition, whether simple or complex
- Any proposition is **logically equivalent** to the disjunction of all atomic events that required the **truth of proposition**.

Prior Probability

- The **unconditional** or **prior probability** associated with a proposition a , is the **degree of belief** according to the absence of any other information;
- it is written as $P(a)$.
- For example, if the prior probability that one have a cavity is 0.1, then we would write
 - $P(\text{Cavity} = \text{true}) = 0.1$ or $P(\text{cavity}) = 0.1$.
- It is important to remember that $P(a)$ can be used only when **there is no other information**.

Prior Probability...

- we will use an expression $P(\textit{Weather})$, which denotes a **vector** of values, for the probabilities of each individual state of the weather.
 - $P(\textit{Weather} = \textit{sunny}) = 0.7$
 - $P(\textit{Weather} = \textit{rain}) = 0.2$
 - $P(\textit{Weather} = \textit{cloudy}) = 0.08$
 - $P(\textit{Weather} = \textit{snow}) = 0.02$.
- we may simply write
- $P(\textit{Weather}) = (0.7, 0.2, 0.08, 0.02)$.
- This statement defines a **prior probability distribution** for the random variable *Weather*.

Conditional Probability

- The **conditional** or **posterior** probabilities notation is $P(a|b)$,
- where a and b are any proposition.
- This is read as "the probability of a , given that *all* we know is b ."
- For example, $P(\text{cavity} | \text{toothache}) = 0.8$
- if a patient is observed to have a toothache and no other information is yet available, then the probability of the patient's having a cavity will be 0.8.

Conditional probabilities...

- Conditional probabilities can be defined in terms of unconditional probabilities.
- The equation is

$$P(a|b) = \frac{P(a \wedge b)}{P(b)}$$

- whenever $P(b) > 0$.
- This equation can also be written as
- $P(a \wedge b) = P(a/b) P(b)$ which is called the **product rule**.

Basic Axioms of Probability

- All probabilities are between 0 and 1. For any proposition a ,

$$0 \leq P(a) \leq 1$$

- Necessarily true (i.e., valid) propositions have probability 1, and necessarily false (i.e., unsatisfiable) propositions have probability 0.

$$P(\text{true}) = 1 \qquad P(\text{false}) = 0 .$$

- The probability of a disjunction is given by

$$P(a \vee b) = P(a) + P(b) - P(a \wedge b) .$$